Week-2

Develop a Java program to create a class Book with members Book ID, Title,Author, and Price. Use a constructor to initialize the details. Include methods to display book information, overload search() to search by Book ID and Title,and create a method that accepts another Book object and returns the costlier book. Use a static member to maintain the total number of books created. In the main class, create multiple book objects and demonstrate all the operations.

```
class Book {
    int bookId;
    String title;
    String author;
    double price;

    // Static variable to count books
    static int count = 0;

    // Parameterized constructor
    Book(int bookId, String title, String author, double price) {
        this.bookId = bookId;
        this.title = title;
        this.author = author;
        this.price = price;
        count++;
    }

    // Display book information
    void display() {
        System.out.println("Book ID: " + bookId);
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("Price: Rs. " + price);
        System.out.println();
    }

    // Search by Book ID
    void search(int id) {
        if (bookId == id) {
            System.out.println("Book found by ID:");
            display();
```

```
        }
    }

    // Overloaded search method: Search by Title
    void search(String title) {
        if (this.title.equalsIgnoreCase(title)) {
            System.out.println("Book found by Title:");
            display();
        }
    }

    // Return the costlier book
    Book costlierBook(Book b) {
        if (this.price >= b.price) {
            return this;
        } else {
            return b;
        }
    }
}

public class Main {
    public static void main(String[] args) {

        // Creating book objects
        Book b1 = new Book(101, "Java", "James", 500);
        Book b2 = new Book(102, "Python", "Guido", 700);
        Book b3 = new Book(103, "C Programming", "Dennis", 400);

        // Display book information
        System.out.println("Book Details:");
        b1.display();
        b2.display();
        b3.display();

        // Search by Book ID
        b1.search(101);

        // Search by Title
        b2.search("Python");
```

```
        // Find the costlier book
        Book costly = b1.costlierBook(b2);
        System.out.println("Costlier Book:");
        costly.display();

        // Display total number of books
        System.out.println("Total Books Created: " + Book.count);

        System.out.println("NAME: Deekshitha H S");
        System.out.println("USN: 1BM26CS605-T");
    }
}
```

OUTPUT

```
----jGRASP exec: java Main
Book Details:
Book ID: 101
Title: Java
Author: James
Price: Rs. 500.0

Book ID: 102
Title: Python
Author: Guido
Price: Rs. 700.0

Book ID: 103
Title: C Programming
Author: Dennis
Price: Rs. 400.0

Book found by ID:
```

```
 ----jGRASP exec: java Main
Book found by ID:
Book ID: 101
Title: Java
Author: James
Price: Rs. 500.0

Book found by Title:
Book ID: 102
Title: Python
Author: Guido
Price: Rs. 700.0

Costlier Book:
Book ID: 102
Title: Python
Author: Guido
Price: Rs. 700.0

Total Books Created: 3
NAME: Deekshitha H S
USN: 1BM26CS605-T

 ----jGRASP: Operation complete.
```